Kasturi Sai Rishi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.03
Narayana Junior College	Class XII (Telangana Board of Intermediate Education)	2023	96.90%
Narayana School	Class X (Board of Secondary Education, Telangana State)	2021	10.00%

EXPERIENCE

Oracle India Private Limited, *Member Technical Staff Intern* May 2026 - Jul 2026

AI-Powered Fault Injection and Automated RCA for Edge-Case Resilience in ADB-S

- Built a **fault injection framework** using **SQL, PL/SQL, Bash, and Python** to reproduce Oracle database hangs, resource contention, and failure scenarios for resilience testing
- Developed a **HANGANALYZE trace analyzer** to reconstruct **blocker-waiter dependency graphs** and automate **Root Cause Analysis (RCA)** with structured diagnostic reports
- Implemented **Oracle RAC resource-pressure workloads** for PGA, shared pool, and parallel execution, with **GV$/V$ monitoring, Prometheus, and Grafana** for system-level analysis

PROJECTS

Cricket Tournament Manager

Collaborated Project - MySQL, SQL, HTML, CSS, JavaScript

- Developed a **full-stack web application** for end-to-end cricket tournament management, supporting **team and player management, match scheduling, real-time match updates, and tournament standings**
- Architected a **relational database** using **MySQL**, designing **ER and UML models** and applying **normalization up to BCNF** to maintain data integrity across tournaments, teams, players, matches, and statistics
- Implemented **optimized SQL queries and stored procedures** for **match updates, schedule generation, player statistics, and points-table tracking**, integrating database operations with the application's frontend and backend modules

Two-Level Parallelized Closeness Centrality

Collaborated Project - C++ / Github

- Implemented **parallel algorithms** for exact **closeness centrality** on large-scale graphs using **BFS, multi-source BFS, OpenMP parallelization, SIMD vectorization, and bitwise SpMM** to accelerate graph traversal and distance computation
- Developed a **Biconnected Component (BCC)**-aware optimization using **Tarjan's algorithm, articulation points, Block-Cut Trees, and dynamic programming** to decompose graphs and reduce redundant BFS computations across structurally connected components
- Optimized **graph analytics and memory utilization** through parallel traversal and component-level computation, enabling efficient computation of **shortest-path distances, node centrality scores, and graph connectivity properties** for large-scale networks

Kademlia Distributed Hash Table File Sharing Pool

Collaborated Project - C++, UDP Sockets / Github

- Implemented a **peer-to-peer file sharing system** in **C++** using **UDP sockets** and the **Kademlia Distributed Hash Table (DHT)** protocol for decentralized file discovery, storage, and retrieval across network peers

- Developed **Kademlia node discovery, XOR-based routing, distributed key-value storage, and iterative lookup** mechanisms to efficiently locate peers and files in a decentralized network without a centralized server
- Implemented **file publishing, retrieval, peer communication, and routing-table management** to support scalable and fault-tolerant P2P communication, gaining hands-on experience with **distributed systems, networking, socket programming, and DHT architecture**

AFT (Another Frequency Thing) Compiler

Collaborated Project - LLVM, C++, Flex, Bison / Github

- Engineered a **statically-typed imperative domain-specific language (DSL)** and compiler toolchain in **C++**, implementing the complete pipeline from **lexical analysis and parsing to AST construction and LLVM IR generation**
- Implemented **lexical analysis using Flex** and a **context-free grammar parser using Bison**, along with an **AST node hierarchy, AST visualization, symbol/scope management, and semantic analysis**
- Integrated the **LLVM API** to build a compiler backend targeting **LLVM Intermediate Representation (IR)**, generating **.ll** files and supporting LLVM versions **16-19** for optimized compilation workflows

RISC-V Cache Simulator

Coursework: Collaborated (team of 2) - Computer Architecture - C++ / Github

- Extended a custom **RISC-V simulator** in **C++** with a configurable **data cache subsystem**, enabling architectural experimentation across different cache organizations, associativities, and memory access patterns
- Implemented **direct-mapped, set-associative, and fully-associative caches** with configurable **LRU, FIFO, and RANDOM replacement policies** and **write-through/write-back strategies**, including tag indexing and cache-line management
- Simulated realistic **cache hit/miss behavior, dirty-bit management, memory updates, and cache-memory consistency**, while generating **per-run cache statistics and execution traces** for performance analysis

SKILLS

Languages: C, C++, CUDA C++, Python (NumPy, pandas, matplotlib), Bash

Parallel/Systems: Multithreading (pthreads), Linux, RISC-V (ISA & Assembly), Verilog

Tools: Git, LaTeX, Vim, MySQL

RELEVANT COURSES

Core CS: Computer Architecture, Data Structures, Operating Systems, Database Management Systems, Compilers, Computer Networks, Algorithms, Software Engineering, Foundations of Machine Learning

Math/CS Theory: Theory of Computation, Discrete Mathematics, Transform Techniques, Linear Algebra, Probability & Random Variables, Convex Optimization, Linear Optimization

Systems & Parallel Computing: Digital Circuits, GPU Programming, Parallel and Concurrent Programming, Quantum Computing

Algorithms & Data: Cryptology, Number Theory & Algebra, Deep Learning, Fraud Analytics, Algorithmic Techniques for Large Graphs

POSITIONS OF RESPONSIBILITY

Academic Affairs Secretary, IIT Hyderabad

- Represented **4,000+ students** in the Senate as part of a **six-member student team**, working with the Director, Dean of Academics, and HoDs. Acted as the **key liaison** between students and the academic office, addressing academic concerns and facilitating communication.

Teaching Assistant - Compilers (CS3320/CS3423)

- Supported **lab sessions** on compiler front-end and back-end development, covering **lexical analysis**, **parsing**.

Teaching Assistant - Theory of Computation (CS2030)

- Evaluated exams covering **finite automata**, **regular languages**, and **context-free grammars**. Guided students through problems involving **Turing machines** and **decidability**.

Teaching Assistant - Computer Architecture (CS2323)

- Assisted in weekly **lab sessions** and doubt-clearing hours, helping students understand **pipelining**, **cache hierarchy**, and **instruction set design**. Guided students through assignments on **datapath** and **control unit design**.

Teaching Assistant - Introduction to Programming (ID1063)

- Assisted in weekly **lab sessions** and biweekly exam clarification sessions, helping students understand core **C programming** concepts including **control structures**, **arrays**, **pointers**, **functions**, and **file operations**.

Hospitality and Logistics Head - DIESTA (Inter-Department Fest), IIT Hyderabad

- Led a team of **15 coordinators** and 2 co-heads, managing **logistics** and **hospitality** for judges and guests across **6+ daily events** over **7 consecutive days**.

EXTRACURRICULAR

Active member of **NSS (National Service Scheme)**, IITH

Engage in **e-gaming** as a recreational activity and participate in occasional competitions

Participating in contests on Codeforces - building algorithmic thinking and problem-solving skills through competitive programming.